

Proof versus Pattern Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	Time	30-40 min
Difficulty	Medium	ID	proof-versus-pattern

Big idea: Examples are not proof.

Overview

Students distinguish empirical pattern spotting from mathematical proof.

Student Prompt

Inspect Proof versus Pattern Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Proof versus Pattern Stimulus A	Student-facing stimulus 1: A number pattern such as $1 + 2 + \dots + n = n(n+1)/2$ that invites both pattern-spotting and proof. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Proof versus Pattern Stimulus B	Student-facing stimulus 2: A visual pattern students extend before testing whether it proves anything. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Proof versus Pattern Reveal / Twist	Student-facing stimulus 3: A comparison between inductive confidence and deductive proof. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Examples are not proof.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Proof versus Pattern Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Proof versus Pattern Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Why is proof valued more than observation in mathematics?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how examples are not proof.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare a starter stimulus using: A number pattern such as $1 + 2 + \dots + n = n(n+1)/2$ that invites both pattern-spotting and proof.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- A number pattern such as $1 + 2 + \dots + n = n(n+1)/2$ that invites both pattern-spotting and proof.
- A visual pattern students extend before testing whether it proves anything.
- A comparison between inductive confidence and deductive proof.

Suggested Timing

Stage	Teacher move	Watch for
1	Students calculate $n^2 + n + 41$ for $n = 0, 1, 2, \dots$ and record whether each output is prime.	Assumptions, confidence shifts, evidence claims, and language choices.
2	They predict whether the expression always gives primes.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Reveal a counterexample, such as $n = 40$.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Students write the difference between many examples and proof.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.
- Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Monty Hall Shock
- Axiom Switch Universe
- Calculator Trust Test